

Substitution and Transposition Ciphers

Information Security Laboratory – Assignment 1

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Aim

To study and implement classical **substitution** and **transposition** cipher techniques, and to perform encryption and decryption using each of them.

1. Substitution techniques

- i. Caesar cipher
- ii. Playfair cipher
- iii. Hill cipher
- iv. Vigenere cipher

2. Transposition techniques

- Rail Fence cipher
- Row & Column (Columnar) Transposition

Theory

Substitution vs. Transposition

A **substitution cipher** replaces each plaintext symbol with another symbol. The *position* of the symbol is kept, but its *identity* is changed.

A **transposition cipher** keeps every plaintext symbol unchanged but rearranges their *positions* according to some system. The letters are the same; only their order changes.

Both are classical (pre-modern) ciphers and are easily broken today, but they form the conceptual foundation of modern symmetric cryptography (confusion and diffusion).

Part 1 – Substitution Techniques

i. Caesar Cipher

Principle

The Caesar cipher shifts every letter of the plaintext by a fixed number k positions down the alphabet (wrapping around).

Encryption: $C = (P + k) \bmod 26$

Decryption: $P = (C - k) \bmod 26$

where P and C are letters mapped to $0..25$ ($A=0, B=1, \dots Z=25$).

Example ($k = 3$)

Plaintext	:	H	E	L	L	O
Values	:	7	4	11	11	14
+3 mod 26	:	10	7	14	14	17
Ciphertext	:	K	H	O	O	R

Encryption: HELLO -> KHOOR

Decryption: KHOOR - 3 (mod 26) -> HELLO

ii. Playfair Cipher

Principle

The Playfair cipher encrypts **digraphs** (pairs of letters) using a 5x5 key matrix built from a keyword (I and J share one cell).

Rules:

1. Build a 5x5 matrix: keyword first (no repeats), then remaining letters (I/J combined).
2. Split the plaintext into letter pairs. If a pair has two identical letters, insert X between them. If the text length is odd, pad with X.
3. For each pair:
 - **Same row** -> replace each letter with the one to its **right**.

- **Same column** -> replace each letter with the one **below** it.
- **Rectangle** -> replace each letter with the letter in its own row but in the column of the other letter.

Decryption reverses the shifts (left / up).

Example (keyword = MONARCHY)

Key matrix:

```
M O N A R
C H Y B D
E F G I/J K
L P Q S T
U V W X Z
```

Encrypt plaintext INSTRUMENTS -> pairs: IN ST RU ME NT SX

```
IN -> GA    (rectangle)
ST -> TL    (same row)
RU -> MZ    (rectangle)
ME -> LC    (same column, wrap)
NT -> AQ    (rectangle)
SX -> BX / handled as pair with padding
```

Encryption: INSTRUMENTS -> GATLMZCLRQXA (pairs shown above)

Decryption applies the inverse (shift left / up / rectangle) to recover the plaintext, then strips the padding X.

iii. Hill Cipher

Principle

The Hill cipher is a **polygraphic** substitution cipher based on linear algebra. Plaintext is grouped into vectors of size n and multiplied by an $n \times n$ key matrix K (mod 26).

Encryption: $C = (K \cdot P) \bmod 26$

Decryption: $P = (K^{-1} \cdot C) \bmod 26$

K must be invertible mod 26 (i.e. $\gcd(\det(K), 26) = 1$).

Example (2x2 key)

```
K = | 3  3 |      Plaintext = "HI"
    | 2  5 |
```

Map letters: H=7, I=8, so $P = [7, 8]$.

Encryption:

```
C0 = (3*7 + 3*8) mod 26 = (21 + 24) mod 26 = 45 mod 26 = 19 -> T
C1 = (2*7 + 5*8) mod 26 = (14 + 40) mod 26 = 54 mod 26 = 2 -> C
```

Encryption: HI -> TC

Decryption: $\det(K) = (35 - 32) = 9$, and $9^{-1} \bmod 26 = 3$.

```
K^-1 = 3 * | 5 -3 | mod 26 = | 15 17 |
           | -2  3 |           | 20  9 |
```

$P_0 = (15 \times 19 + 17 \times 2) \bmod 26 = (285 + 34) \bmod 26 = 319 \bmod 26 = 7 \rightarrow H$
 $P_1 = (20 \times 19 + 9 \times 2) \bmod 26 = (380 + 18) \bmod 26 = 398 \bmod 26 = 8 \rightarrow I$
 TC $\rightarrow$ HI.

iv. Vigenere Cipher

Principle

The Vigenere cipher is a **poly-alphabetic** substitution cipher. A repeating keyword defines a different Caesar shift for each position.

Encryption: $C = (P + K) \bmod 26$

Decryption: $P = (C - K) \bmod 26$

where K is the keyword letter (repeated) aligned with each plaintext letter.

Example (keyword = KEY)

Plaintext : H E L L O
 Key (rep) : K E Y K E
 P values : 7 4 11 11 14
 K values : 10 4 24 10 4
 Sum mod 26: 17 8 9 21 18
 Ciphertext: R I J V S

Encryption: HELLO $\rightarrow$ RIJVS

Decryption: RIJVS - KEYKE (mod 26) $\rightarrow$ HELLO

Part 2 – Transposition Techniques

Rail Fence Cipher

Principle

Plaintext is written diagonally over a number of “rails” (rows) in a zig-zag pattern, then read off row by row to form the ciphertext.

Example (rails = 3)

Plaintext: WEAREDISCOVERED

Write in zig-zag over 3 rails:

W . . . E . . . C . . . R . .
 . E . R . D . S . O . E . E .
 . . A . . . I . . . V . . . D

Read row by row:

Row 1: W E C R
 Row 2: E R D S O E E
 Row 3: A I V D

Encryption: WEAREDISCOVERED $\rightarrow$ WECRERDSOEAAIVD

Decryption: rebuild the zig-zag frame of the same size, place the ciphertext into the rails row by row, then read diagonally to recover WEAREDISCOVERED.

Row & Column (Columnar) Transposition

Principle

Write the plaintext in rows under a keyword. Number the columns by the alphabetical order of the keyword letters. Read the columns out in that numeric order to produce the ciphertext.

Example (keyword = ZEBRA)

Column order from ZEBRA (alphabetical rank): Z=5 E=2 B=1 R=4 A=... Ranks: A=1, B=2, E=3, R=4, Z=5 -> keyword columns get order Z->5, E->3, B->2, R->4, A->1.

Plaintext: MEETMEATNOON (pad with X to fill the grid)

```
Keyword : Z  E  B  R  A
Order   : 5  3  2  4  1
-----
      M  E  E  T  M
      E  A  T  N  O
      O  N  X  X  X
```

Read columns in order 1..5:

```
Order 1 (A): M O X
Order 2 (B): E T X
Order 3 (E): E A N
Order 4 (R): T N X
Order 5 (Z): M E O
```

Encryption: MEETMEATNOON -> MOXETXEANTNXMEO

Decryption: recreate the grid dimensions, fill the columns back in numeric order, then read row by row and strip the padding to recover MEETMEATNOON.

Result

The substitution ciphers (Caesar, Playfair, Hill, Vigenere) and the transposition ciphers (Rail Fence, Columnar) were studied and their encryption / decryption processes were successfully demonstrated with worked examples.

Technique	Type	Unit encrypted
Caesar	Substitution	single letter
Playfair	Substitution	digraph (2 letters)
Hill	Substitution	n-letter block
Vigenere	Substitution	single letter
Rail Fence	Transposition	reorders letters
Columnar	Transposition	reorders letters